

Lesson 5

# Building an Effective Power BI Dashboard

How to plan, structure, and build a Power BI dashboard  
by following a step-by-step design process

March 2025

# **How to build a good dashboard?**

*Key Steps and Best Practices for Effective Design.*

# What makes a good dashboard?

Key Steps and Best Practices for Effective Design.

A dashboard is a visual interface that displays key metrics and insights in an easy-to-understand format, enabling users to make data-driven decisions.

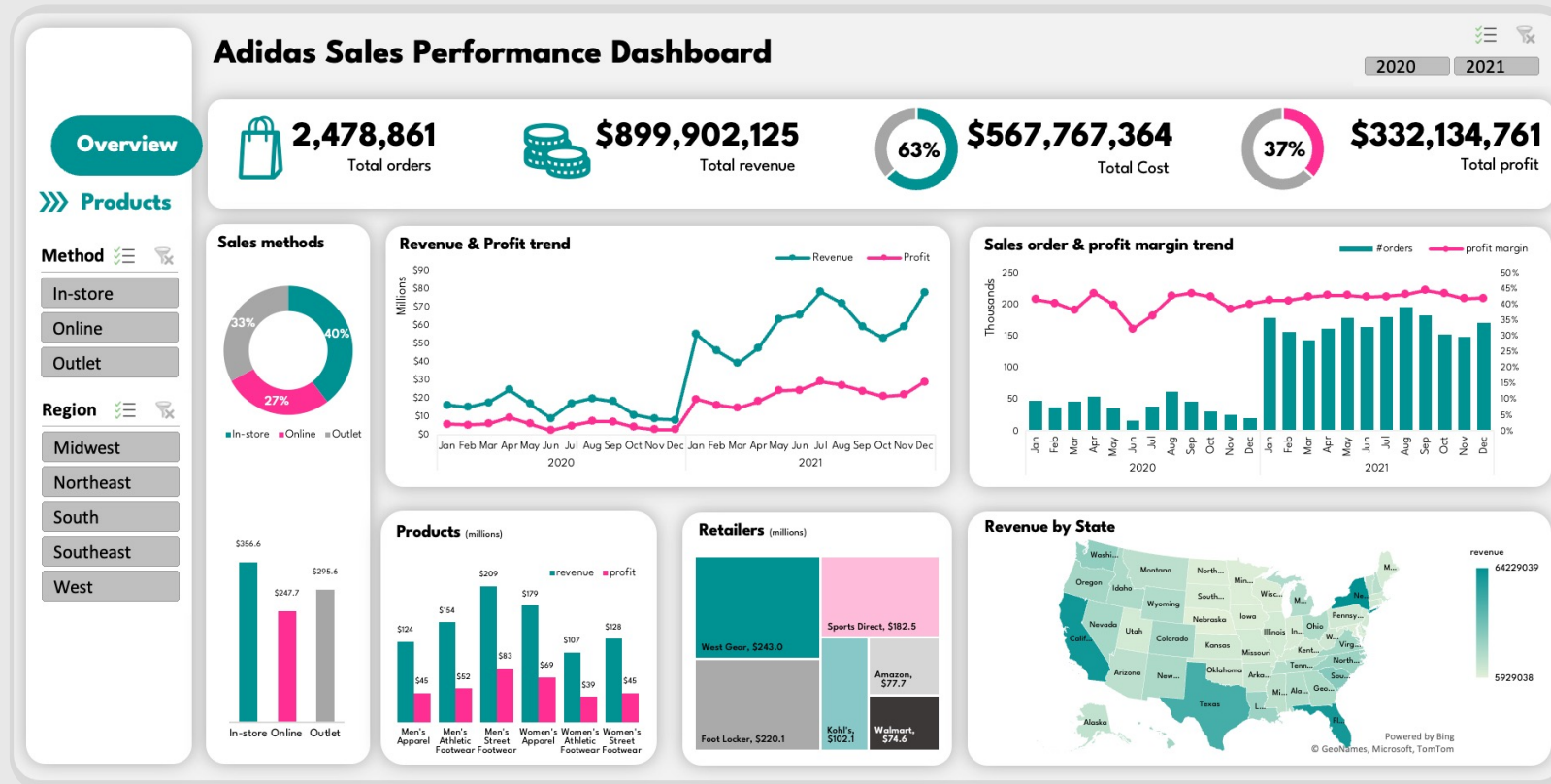
Interactivity

Design

Clarity

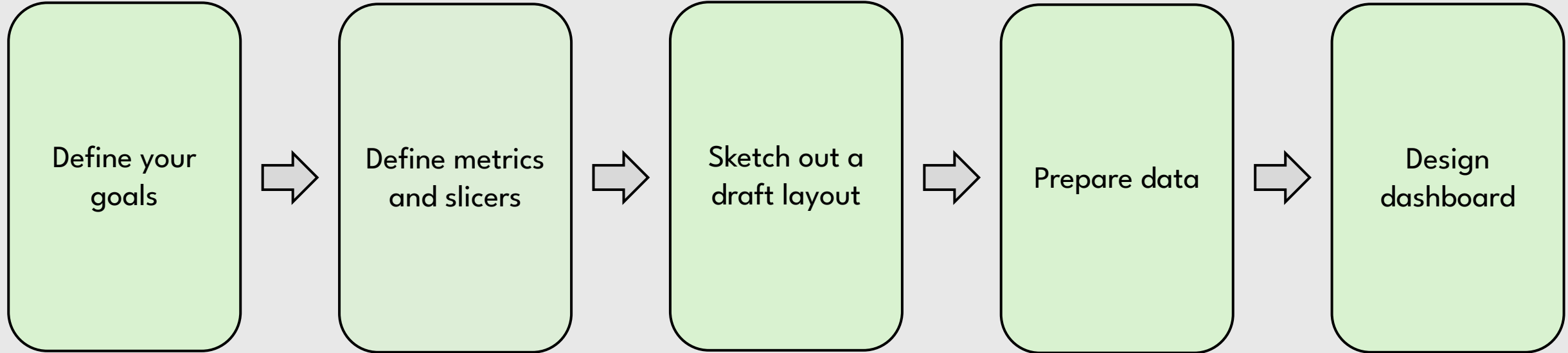
Informative

Actionable  
Insights



# Steps to Build a Good Dashboard in Power BI?

*Key Steps and Best Practices for Effective Design.*



## **1 Define your goals.**

*A Beginner's Guide to build an effective dashboard.*

A well-designed Power BI dashboard starts with a clear understanding of its purpose, the target users, and the type of dashboard needed. This ensures that the data is presented effectively and meets business needs.

# Define your goals.

*A Beginner's Guide to build an effective dashboard.*



## The Purposes:

What is the goal of the dashboard

- What key business decisions will it support?
- What questions should the dashboard answer?
- How will success be measured?



## The End Users :

Who will use the dashboard

- What level of detail do they need?  
(High-level summary vs. detailed breakdown?)
- How frequently will they use it?  
(Daily, Weekly, Monthly, On-Demand?)
- What is their technical expertise?  
(Basic viewers vs. Data Analysts?)



## Type of Dashboard:

Best fits the purpose

- Operational, Strategic, or Analytical dashboard?

# Define the purposes?

End Users & Purpose & Type of Dashboard.

## Key questions to answer:

- 1 What is the goal of the dashboard? (Why are we building it?)
- 2 What key business decisions will it support?
- 3 What questions should the dashboard answer?
- 4 How will success be measured?



| Business Problem                        | Purpose of Dashboard                    |
|-----------------------------------------|-----------------------------------------|
| Sales are declining in certain regions  | Identify sales trends & revenue sources |
| Customer churn is high                  | Track retention & loyalty metrics       |
| Marketing campaigns have low conversion | Measure campaign performance            |
| Inventory stockouts are causing delays  | Optimize supply chain monitoring        |

# Define the end users?

End Users & Purpose & Type of Dashboard.

## Key questions to answer:

- 1 Who will use the dashboard? (Executives, Sales, Marketing, Operations, Finance?)
- 2 What level of detail do they need? (High-level summary vs. detailed breakdown?)
- 3 How frequently will they use it? (Daily, Weekly, Monthly, On-Demand?)
- 4 What is their technical expertise? (Basic viewers vs. Data Analysts?)



| User               | Dashboard Needs                                       | Preferred Visuals             |
|--------------------|-------------------------------------------------------|-------------------------------|
| CEO                | High-level company revenue, profit, growth trends     | KPI Cards, Line Charts        |
| Sales Manager      | Order trends, top-performing products, regional sales | Bar Charts, Maps, Slicers     |
| Marketing Analyst  | Campaign performance, website traffic, ROI analysis   | Funnel Charts, Pie Charts     |
| Finance Controller | Monthly budget, cash flow, cost analysis              | Table Views, KPIs, Bar Charts |

# Define the type of dashboard?

End Users & Purpose & Type of Dashboard.

## Key questions to answer:

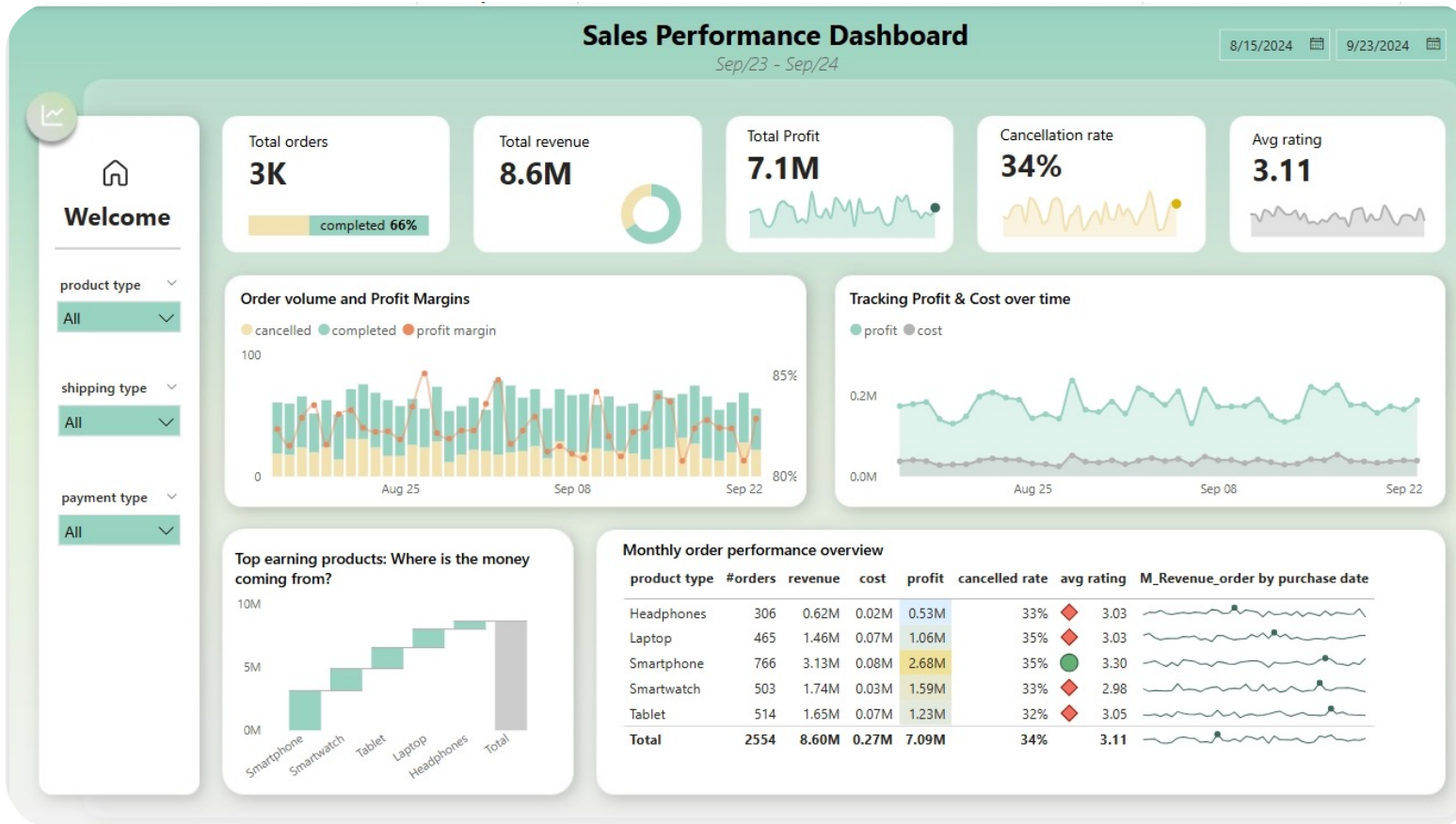
1 What kind of dashboard best fits the purpose and users?

| Dashboard Type        | Purpose                                | Examples                                     | Users                    |
|-----------------------|----------------------------------------|----------------------------------------------|--------------------------|
| Operational Dashboard | Real-time monitoring & daily tracking  | Live sales updates, website traffic          | Sales Teams, Support     |
| Strategic Dashboard   | High-level KPIs for decision-making    | Quarterly financial overview, executive KPIs | C-Level Executives       |
| Analytical Dashboard  | Deep data exploration & trend analysis | Customer segmentation, churn prediction      | Data Analysts, Marketing |



# Operational Dashboard.

Real-Time Monitoring & Daily Tracking



## Purposes:

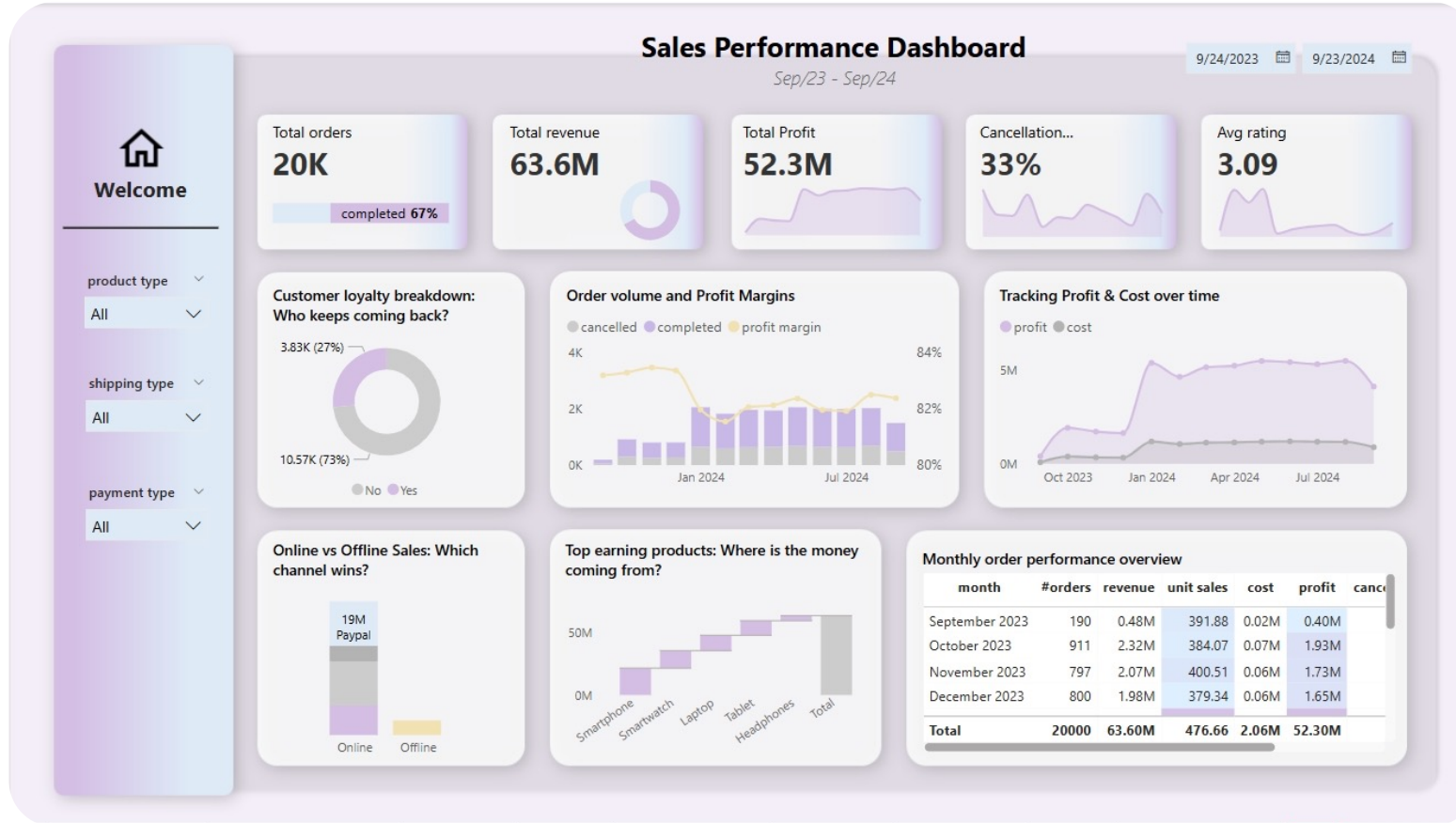
- Focuses on day-to-day operations and real-time updates.
- Monitors ongoing performance metrics, helping users react quickly.

## Key features:

- Live Data Refresh: Frequent updates (hourly, real-time).
- Performance Metrics: Displays short-term KPIs
- Simple Visuals: Uses cards, tables, and status indicators for quick interpretation.

# Strategic Dashboard.

High-Level KPIs for Decision-Making



## Purposes:

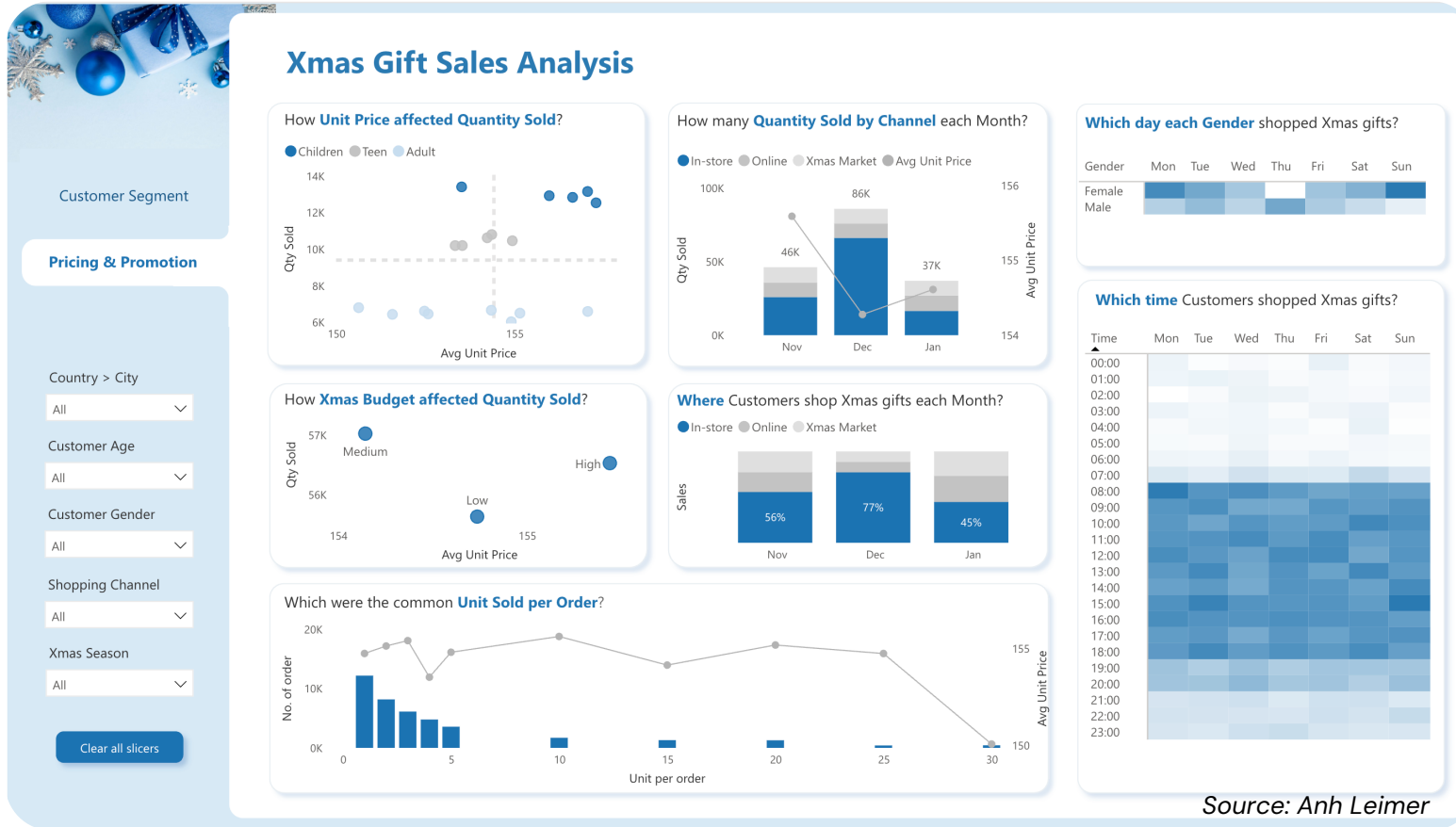
- Designed for executives & senior management to track long-term business performance.
- Focuses on big-picture KPIs rather than real-time updates.

## Key features:

- Historical Trends: Year-over-year (YoY) growth, profitability, and cost analysis.
- Comparisons & Benchmarks: Compares performance across business units, regions, or teams.
- Key Business Indicators (KPIs): Revenue, profit margins, growth %, market share.

# Analytical Dashboard.

High-Level KPIs for Decision-Making



## Purposes:

- Focuses on day-to-day operations and real-time updates.
- Monitors ongoing performance metrics, helping users react quickly.

## Key features:

- Live Data Refresh: Frequent updates (hourly, real-time).
- Performance Metrics: Displays short-term KPIs
- Simple Visuals: Uses cards, tables, and status indicators for quick interpretation.

## 2 Define Metrics and Slicers.

*A Beginner's Guide to build an effective dashboard.*

Choosing the **right metrics (KPIs) and slicers (filters)** ensures that a dashboard provides actionable insights and allows users to explore data effectively.

# Define Metrics and Slicers.

*A Beginner's Guide to build an effective dashboard.*



## The Right Metrics:

How to choose the right metrics?

- KPIs should be relevant to business goals.
- Use a mix of leading & lagging indicators.
- Metrics should be measurable, comparable, and meaningful.



## Defining Slicers (Filters):

For Data Exploration

- Slicers allow users to customize views dynamically.
- Choose slicers based on business needs and report usability.
- Too many slicers make a dashboard complex, too few limit flexibility.

# Steps 2: Identify Key Metrics and KPIs

*Effective guide.*

## 1. Understand the Business Objectives:

- Key Question: **What are the specific goals or challenges the dashboard is designed to address?**
- KPIs should directly align with the broader business objectives.

Examples:

- Increase sales revenue.
- Improve customer retention.
- Reduce operational costs.



# Steps 2: Identify Key Metrics and KPIs

*Effective guide.*

## 2. Differentiate between metrics and KPIs:

- **Metrics:** Quantifiable measures used to track performance (e.g., total sales, website visits).
- **KPIs** (Key Performance Indicators): Strategic metrics that directly measure progress toward business objectives.

### Example:

- Metric: Number of website visitors.
- KPI: Conversion rate (website visitors who made a purchase).



# Sales metrics

## Revenue Metrics:

- ✓ Total Revenue: The total income generated from sales during a specific period.
- ✓ Sales Growth Rate: The percentage increase or decrease in sales over a specific period.
- ✓ Monthly Recurring Revenue (MRR): Monthly revenue from subscriptions.
- ✓ Annual Recurring Revenue (ARR): Yearly revenue from recurring subscriptions.

## Profitability Metrics:

- ✓ Gross Profit Margin: Percentage of revenue remaining after deducting the cost of goods sold (COGS).
- ✓ Net Profit Margin: Percentage of revenue remaining after all expenses.

## Performance Metrics:

- ✓ Average Order Value (AOV): Average revenue per order.
- ✓ Sales Conversion Rate: Percentage of leads or opportunities converted into sales.
- ✓ Win Rate: Percentage of sales opportunities successfully closed.

## Customer Metrics:

- ✓ Customer Retention Rate: Percentage of customers retained over time.
- ✓ Churn Rate: Percentage of customers lost during a specific period.
- ✓ Upsell/Cross-Sell Revenue: Additional revenue generated from existing customers.



# Marketing performance metrics

## Digital Marketing Metrics:

- ✓ Website Traffic: Total number of visitors to the website.
- ✓ Bounce Rate: Percentage of visitors who leave after viewing one page.
- ✓ Click-Through Rate (CTR): Percentage of clicks from total ad impressions.
- ✓ Conversion Rate: Percentage of users completing desired actions.

## Campaign Performance:

- ✓ Campaign ROI: Return on investment for a specific marketing campaign.
- ✓ Cost Per Acquisition (CPA): Cost required to acquire a new customer.
- ✓ Cost Per Lead (CPL): Cost of generating a single lead.

## Social Media Metrics:

- ✓ Engagement Rate: Level of interaction (likes, shares, comments) on social media.
- ✓ Follower Growth Rate: Percentage increase in followers over time.
- ✓ Reach: Number of unique users who viewed content.

## Brand Metrics:

- ✓ Brand Awareness: Extent to which a brand is recognized by its audience.
- ✓ Customer Sentiment Analysis: Analysis of customer emotions toward the brand.



# Operations metrics

## Efficiency Metrics:

- ✓ Order Fulfillment Time: Time taken to process and deliver an order.
- ✓ Process Cycle Efficiency (PCE): Percentage of process time adding value.
- ✓ On-Time Delivery Rate: Percentage of orders delivered on or before the promised date.

## Productivity Metrics:

- ✓ Units Produced per Hour: Average units produced per employee/machine per hour.
- ✓ Labor Utilization Rate: Percentage of employee hours used productively.

## Quality Metrics:

- ✓ Defect Rate: Percentage of defective products in total production.
- ✓ Yield Percentage: Percentage of products produced without defects.

## Cost Metrics:

- ✓ Cost Per Unit: Average cost to produce a single unit.
- ✓ Operational Cost Percentage of Revenue: Portion of revenue spent on operations.



# Financial metrics

## Profitability Metrics:

- ✓ Gross Profit Margin: Percentage of revenue remaining after deducting COGS.
- ✓ Operating Profit Margin: Percentage of revenue remaining after operating costs.

## Liquidity Metrics:

- ✓ Current Ratio: Ability to cover short-term liabilities ( $\text{Current Assets} / \text{Current Liabilities}$ ).
- ✓ Quick Ratio: Liquidity excluding inventory ( $\text{Quick Assets} / \text{Current Liabilities}$ ).

## Efficiency Metrics:

- ✓ Return on Investment (ROI): Profitability of an investment ( $\text{Net Profit} / \text{Investment Cost}$ ).
- ✓ Return on Equity (ROE): Profit generated per dollar of equity.

## Growth Metrics:

- ✓ Revenue Growth Rate: Percentage increase in revenue over time.
- ✓ Earnings Per Share (EPS): Profit allocated to each outstanding share.



# HR Management KPIs

## Recruitment Metrics:

- ✓ Time to Hire: Time taken to fill an open position.
- ✓ Cost Per Hire: Total cost incurred to hire a new employee.

## Employee Metrics:

- ✓ Employee Retention Rate: Percentage of employees retained over time.
- ✓ Employee Turnover Rate: Percentage of employees leaving during a specific period.

## Training & Development Metrics:

- ✓ Training Completion Rate: Percentage of employees completing assigned training.
- ✓ Average Training Cost per Employee: Cost incurred for training each employee.

## Performance Metrics:

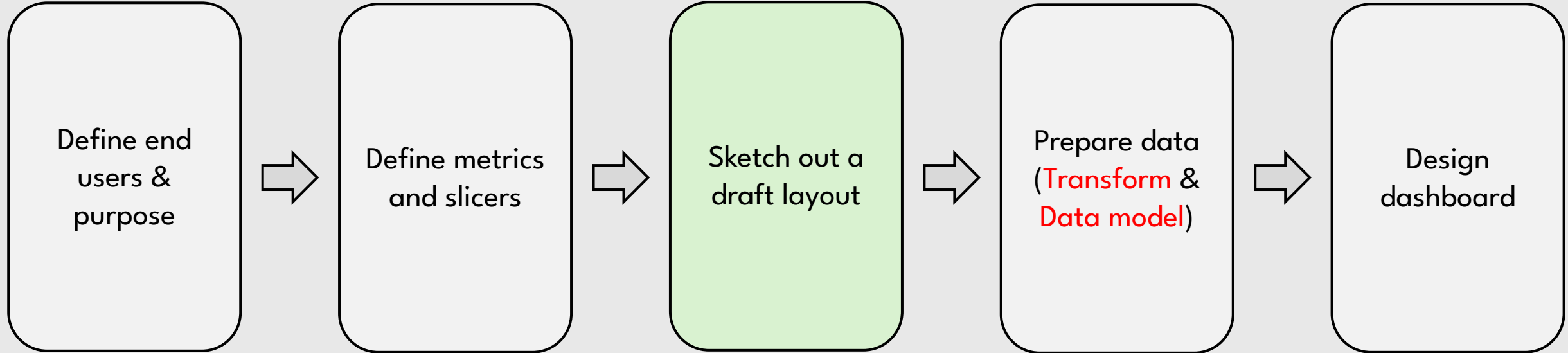
- ✓ Revenue Per Employee: Revenue generated per employee.
- ✓ Internal Promotion Rate: Percentage of job openings filled internally.



[Business Intelligence KIPs Handbook](#)

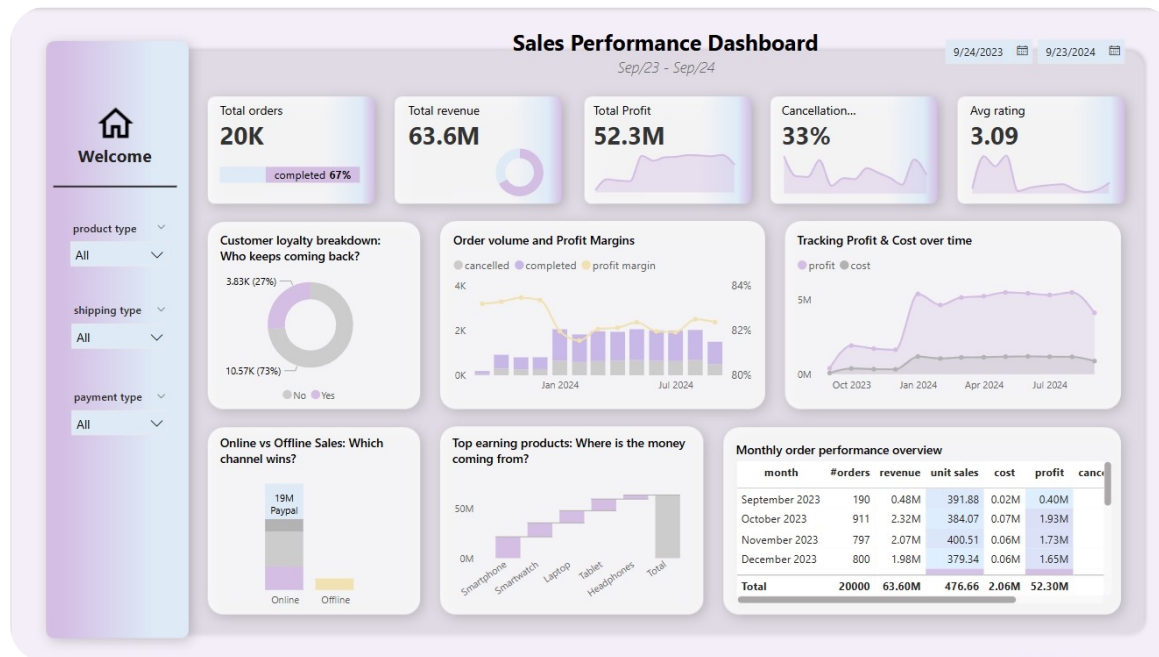
# Steps to Build a Good Dashboard in Power BI?

*Key Steps and Best Practices for Effective Design.*



# Checklist for Dashboard Layout

A dashboard is a visual interface that displays key metrics and insights in an easy-to-understand format, enabling users to make data-driven decisions.



- Critical metrics (KPIs) are placed at the top.
- Related visuals are grouped and aligned.
- The layout follows a logical flow (F-pattern or Z-pattern).
- Responsive for different devices (desktop and mobile).
- Whitespace is used effectively to avoid clutter.
- Design elements are consistent (fonts, colors, chart types).
- Interactivity is intuitive and enhances exploration.
- Feedback is collected, and the layout is refined based on user input.

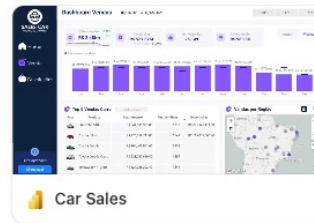
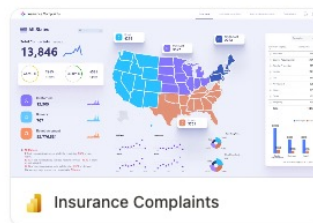
# Power BI Dashboard template

A dashboard is a visual interface that displays key metrics and insights in an easy-to-understand format, enabling users to make data-driven decisions.

## 100+ Power BI Dashboards and Reports Collection

A collection of some of the best Power BI Dashboards & Reports available on the internet

Gallery List



## 2 What do you want to **COMMUNICATE**?

### COMPARISON



Used to **compare values over time or across categories**

#### Common visuals:

- Column/Bar Chart
- Clustered Column/Bar
- Data Table/Heat Map
- Radar Chart
- Line Chart (*time series*)
- Area Chart (*time series*)

### COMPOSITION



Used to **break down the component parts of a whole**

#### Common visuals:

- Stacked Bar/Column Chart
- Pie/Donut Chart
- Stacked Area (*time series*)
- Waterfall Chart (*gains/losses*)
- Funnel Chart (*stages*)
- Tree Map/sunburst (*hierarchies*)

### DISTRIBUTION



Used to **show the frequency of values within a series**

#### Common visuals:

- Histogram
- Density Plot
- Box & Whisker
- Scatter Plot
- Data Table/Heat Map
- Map/Choropleth (*geospatial*)

### RELATIONSHIP



Used to **show correlation between multiple variables**

#### Common visuals:

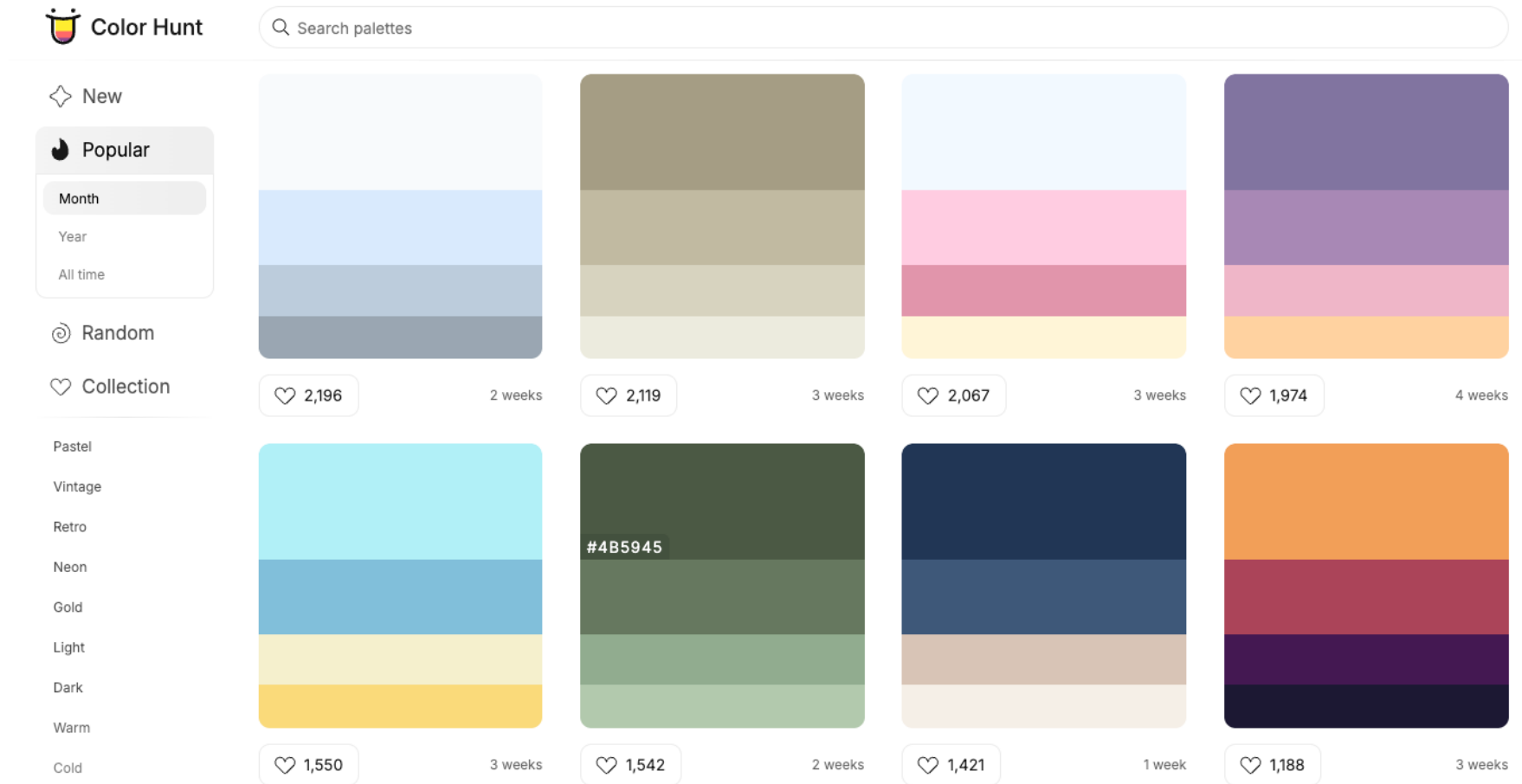
- Scatter Plot
- Bubble Chart
- Data Table/Heat Map
- Correlation Matrix

**Keep it simple!** While there are *hundreds* of charts to choose from, basic options like bars and columns, line charts, histograms and scatterplots often tell the simplest and clearest story

[How to choose a right chart?](#)

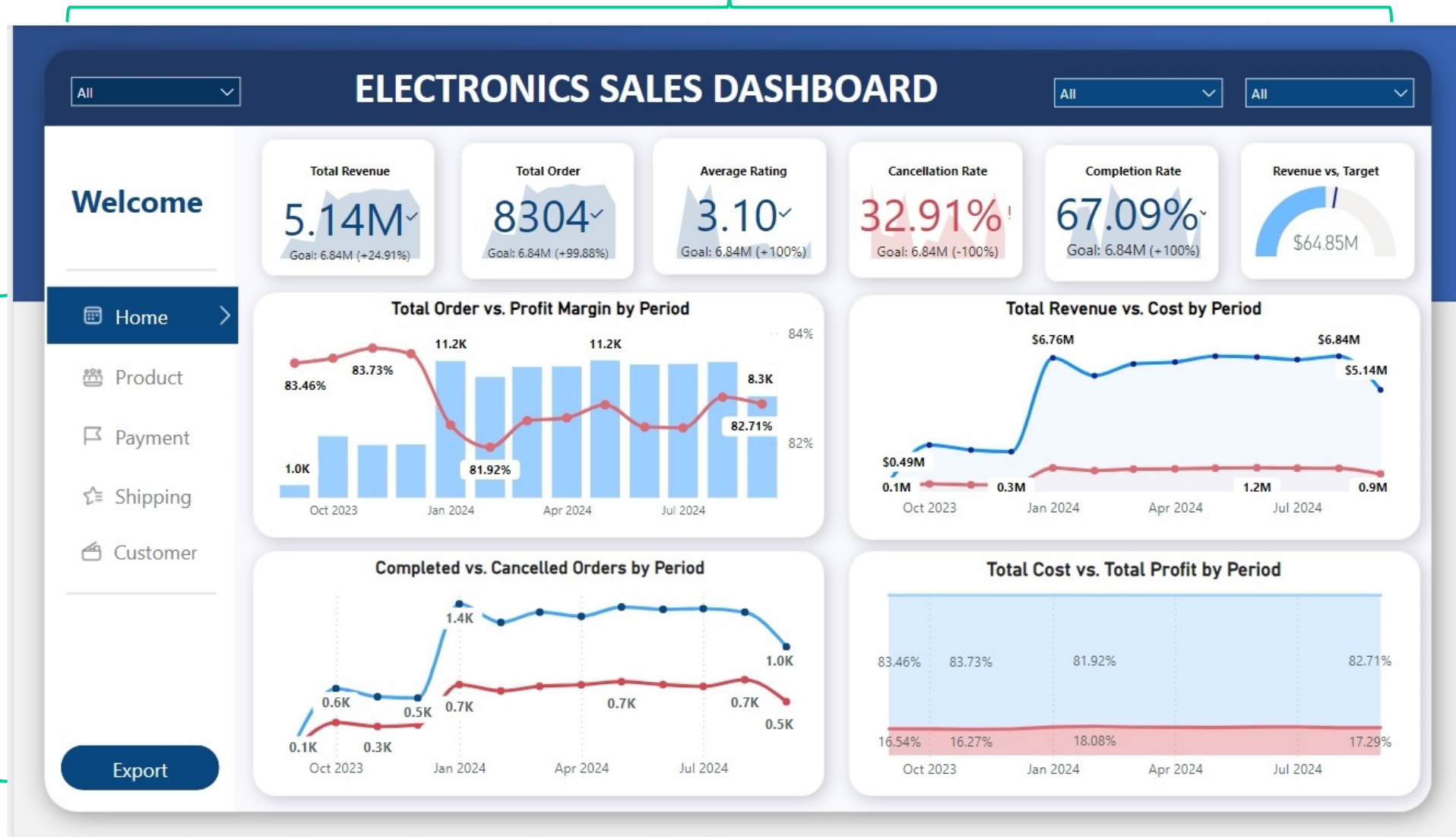
# Color pattern

A dashboard is a visual interface that displays key metrics and insights in an easy-to-understand format, enabling users to make data-driven decisions.



# How to build a good dashboard?

Header



Navigation

# Why Chart Selection Matters

The difference that matters:

## Knowing Power BI

- Creating visualizations
- Connecting to data sources
- Using built-in functions

## Designing Effective Dashboards

Choosing charts that answer questions

Building insight hierarchies

How do we know the charts we choose are right and enough?

*Effective dashboards start with the right chart selection*

# The Question → Metric → Chart Framework

Concept:

Every chart should answer a specific business question



| Question                           | Metric                  | Suitable Chart       |
|------------------------------------|-------------------------|----------------------|
| Is revenue growing?                | Revenue by day/month    | Line chart           |
| Which product performs best?       | Profit by product type  | Bar chart            |
| Where are cancellations happening? | Orders by shipping type | Stacked column chart |
| Are loyal customers more valuable? | Revenue by loyalty flag | KPI + Bar chart      |

If a chart doesn't answer a clear question — it should be removed

# Insight Pyramid – 3 Levels of Analysis

## Concept:

Structure dashboards with 3 levels of insight

Chart Types:  
KPIs  
Cards

**LEVEL 1: OVERVIEW**  
High-level performance at a glance

Chart Types:  
Bar Charts  
Line Charts

**LEVEL 2: BREAKDOWN**  
Performance by key dimensions

Chart Types:  
Scatter Plots  
Heat Maps

**LEVEL 3: DIAGNOSTIC**  
Detailed analysis of specific issues

**Most dashboards only need 4-7 well-chosen visuals**

Focus on making fewer, higher-quality visuals that tell a cohesive story

# MECE Principle + Diagnostic Completeness

## MECE Principle

Mutually Exclusive, Collectively Exhaustive

### Mutually Exclusive

Avoid redundant charts showing the same data

### Collectively Exhaustive

Cover all important dimensions

## Diagnostic Completeness

For each key performance metric

- Trend over time
- Breakdown by key dimensions  
(product, channel, customer)
- Efficiency indicators  
(profit margin, conversion rate)

## Common Incomplete Dashboard Patterns

Only showing revenue without profit or margin

Showing revenue over time but not by product

Key Dimensions to Cover in Most Business Dashboards:

Time • Category • Channel • Customer • Geography

Ask: "If I were making decisions based only on this dashboard, what would I be missing?"

# Example: Incomplete Dashboard Analysis

## E-Commerce Dashboard - Monthly Overview

Revenue Over Time



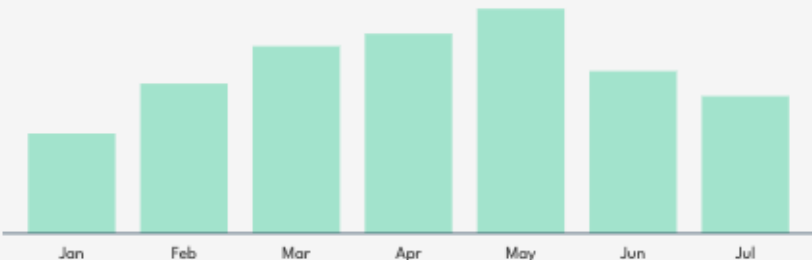
Total Revenue

\$856,421

Total Orders

12,482

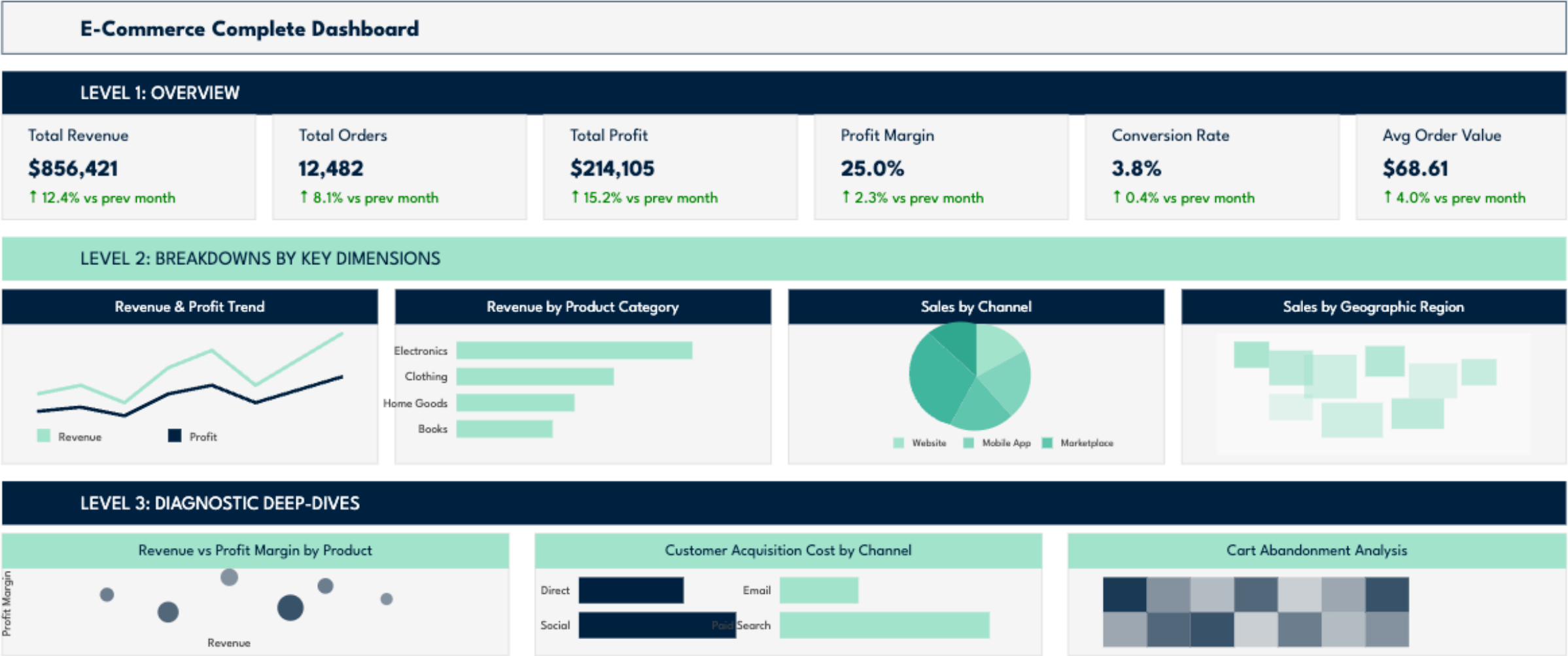
Website Visitors



Why This Dashboard Is Incomplete

# Example: Complete Dashboard Analysis

## E-commerce Dashboard: Properly Structured Insights



# Dashboard Evolution: Incomplete → Complete

Moving from "What Happened" to "Why It Happened and What's Next"

| INCOMPLETE                                                                                                                                                                                                        | COMPLETE                                                                                                                                                                                                                  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div><div>Time Dimension</div><div><div>✓ Revenue over time</div><div>✗ No profit trend comparison</div></div></div>                                                                                              | <div><div>Time Dimension</div><div><div>✓ Revenue over time</div><div>✓ Profit trend alongside revenue</div></div></div>                                                                                                  |
| <div><div>Key Dimensions</div><div><div>✗ No product categories</div><div>✗ No customer segments</div><div>✗ No geographic breakdown</div></div></div>                                                            | <div><div>Key Dimensions</div><div><div>✓ Revenue by product category</div><div>✓ Sales by channel</div><div>✓ Geographic distribution</div></div></div>                                                                  |
| <div><div>Efficiency Indicators</div><div><div>✗ No profit or margin metrics</div><div>✗ No customer acquisition cost</div><div>✗ No conversion rate metrics</div><div>✗ No average order value</div></div></div> | <div><div>Efficiency Indicators</div><div><div>✓ Profit and margin metrics</div><div>✓ Customer acquisition cost by channel</div><div>✓ Conversion rate tracking</div><div>✓ Average order value trends</div></div></div> |

Complete dashboards enable decision makers to understand both what happened AND why

Copyright ©mazedata 2025

# Final Checklist for Chart Validation

Use this checklist to evaluate every chart in your dashboard

- ☐ Does this chart answer a specific question?
- ☐ Does it belong to a clear insight level?  
(Overview, Breakdown, or Diagnostic)
- ☐ Is it redundant or necessary?
- ☐ Does it help decision-makers?
- ☐ Have you gathered stakeholder feedback?

*"The goal isn't creating many charts,  
but the right charts that drive better decisions."*

# Requirements

Xây dựng 1 báo cáo nhằm theo dõi tình hình kinh doanh của công ty theo thời gian, sản phẩm, phương thức giao hàng

